

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 4B

1. Notes: Intro to DTFS

This is not a question, but a brief intro/overview of DTFS

A discrete-time signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ is p -periodic, for an integer $p > 0$, if

$$x[n + p] = x[n] \quad \text{for all } n \in \mathbb{Z}.$$

The *fundamental period* is the smallest such p , and the *fundamental frequency* is $\omega_0 = 2\pi/p$.

Any p -periodic signal can be written as a linear combination of p complex exponentials, given by the *DTFS Synthesis Equation*:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n}, \quad \text{for all } n \in \mathbb{Z}.$$

The p coefficients X_k —called the *DTFS coefficients* of x —are unique.

The index k usually runs from 0 to $p - 1$. It may instead run over any set $\langle p \rangle$ of p consecutive integers, such as $[1, p]$ or $[-1, p - 2]$.

The DTFS coefficients describe the signal in the frequency domain. The magnitude $|X_k|$ tells you how much of the frequency $k\omega_0$ is present in x , and the phase of X_k gives the phase of that component. The coefficients are given by the *DTFS Analysis Equation*:

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}, \quad k = 0, 1, \dots, p - 1.$$

2. Proving Cool DTFS Properties

Given a periodic signal x with period p and frequency $\omega_0 = \frac{2\pi}{p}$, recall the two golden equations for DTFS:

- Synthesis equation:

$$x[n] = \sum_{k \in \langle p \rangle} X_k e^{ik\omega_0 n}$$

- Analysis equation:

$$X_k = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n}$$

Prove the following properties by "mathematically manipulating" these equations:

- (a) Prove that $X_k = X_{-k}^*$ if and only if x is a real-valued signal.

*Note: this is often called the **Conjugate Symmetry** property.*

- (b) Let x be a discrete-time periodic signal with DTFS coefficients X_k . Let $y[n] = x[n - n_0]$ be a time-shifted version of x with DTFS coefficients Y_k . Prove that $Y_k = e^{-jk\omega_0 n_0} X_k$.

*Note: this is often called the **Time Shift** property.*

3. DTFS Analysis of a Simple Signal

Let $x[n]$ be the periodic signal with one period given by $x = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$, i.e. $x[0] = 2, x[1] = 1, x[2] = 0, x[3] = 1$, repeating with period 4.

- (a) Find the DTFS coefficients X_k for $k = 0, 1, 2, 3$. Then, using these coefficients, write $x[n]$ in the form $A + B \cos(\omega_0 n)$.

- (b) Let $y[n] = x[n - 1]$ be a time-shifted version of $x[n]$. Find the DTFS coefficients Y_k of $y[n]$.